Codebook Quantization Compiles a Reservoir into a Finite-State Automaton

A Preprint

Caitlyn Meeks
AuxiLab Tenerife — research@auxi.cafe

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registered 2026-08-11 · internal run log #16–#17 · 2 seeds × 5 codebook sizes, zero misses · replicated before publication · promoted 2026-08-11

Abstract

Can a continuous-state reservoir be *compiled* into a finite-state automaton at negligible cost? An earlier study in this research program asked the question with the obvious instrument — quantizing every coordinate of the state to b bits — and obtained a clean negative: determinism never appeared at any resolution from 1 to 23 bits, because a coarser grid still leaves astronomically many representable states. This study changes one thing: instead of rounding coordinates independently, the state is snapped, at every step, to its nearest vector in a **fixed codebook of K states the unquantized reservoir actually visited** — a hard cap on the cardinality of the state vector itself. That one change reverses the result. (1) At *every* tested K ∈ {16, 64, 256, 1024, 4096}, on both seeds — 10 of 10 cells — the system becomes deterministic and compiles to a finite-state automaton: identical recent-character suffixes of depth 7–11 map to the identical state at every one of 40,000 probed positions. (2) The dial is smooth: prediction cost degrades monotonically as the codebook shrinks, 8/8 adjacent comparisons, with no discontinuity and no reversal. (3) The largest tested codebook is nearly free: K=4096 costs only 0.053–0.057 bits over the unquantized float32 reference — the cheapest tested point on the dial is also the one that compiles.

Keywords reservoir computing · vector quantization · finite-state automata · state-space discretization · determinism · pre-registration

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§1 The question

A reservoir's state is a point in a 1024-dimensional continuum; its dynamics can be checked only statistically. A finite-state machine, by contrast, can be tabulated, audited, and proved things about. An earlier study in this program (internal run log #7) posed the compilation question with per-coordinate state quantization and obtained a striking negative: bits-per-character (bpc) cost degraded smoothly below b≈6 bits and was *exact* to float32 from b=8 — but short-memory determinism stayed at exactly 0.0 at every b from 1 to 23. Rounding re-injects a small perturbation at each step, and a b-bit grid over 1024 coordinates still admits vastly more representable states than any run visits; no state recurred among the 40,000 sampled states. A companion study of normalized fixed-stride descent had meanwhile left an untried construction in the program's notes: *a reservoir whose state is confined to a fixed finite set of anchor states*. This study builds that reservoir and asks whether a hard cardinality cap — rather than a finer or coarser grid — is what forces a reservoir to become a machine.

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§2 Method

The base system is the program's reference sparse-tanh echo state network (ESN) at its best measured operating point (N=1024, spectral radius ρ=0.8, sparse fan-in-10 mixing, the program's standard input map, leak 1.0) [3] — the same anchor cell the earlier per-coordinate quantization study used. One line is added to its update:

```
# C = codebook: K states the unquantized reservoir visited
#     during a fixed 100k-char warmup, sampled once,
#     held fixed for the whole run (no k-means, no synthesis)
x ← argmin_{c ∈ C} ‖x − c‖   # after every update step
```

The snapped state feeds the next step, so the closed-loop dynamics live entirely on the K codebook states — nearest-neighbor vector quantization [1] of the state, with the codebook drawn from the reservoir's own visited set rather than fitted to it. K sweeps {16, 64, 256, 1024, 4096}. Instruments are copied verbatim from the earlier quantization study for direct comparability: ridge and logistic validation bpc on standard text8 splits [4] (2M training characters), and the suffix-determinism probe (T5 in the program's internal instrument numbering): over 40,000 sampled positions, states are grouped by their length-k character suffix, and the probe reports the fraction of multi-sample suffix classes whose members all occupy exactly one state. The freeze depth k* is the smallest depth at which that fraction reaches 1.0 — that is, the shortest window of recent input that fixes the state exactly. A system with finite k* is said to *freeze*: its closed-loop dynamics are then a finite-state automaton over the codebook. k* is judged only where at least 20 multi-sample classes support it — at K=4096 (seed 0), k*=10 rests on 2,172 such classes, not a thin slice. A consistency-check cell (K=∞, no snapping) reproduced the plain unquantized reservoir bit-for-bit through the same code path in both pre-registration pilot runs.

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§3 Pre-registered predictions

- **M1** (existence, per K, judged independently): does k* come out finite at each of the five K's? Named alternative, per K: the snap-to-nearest correction re-injects noise the way rounding did, and a small *cardinality* cap does not force determinism by itself. Freezing at any K is a substantive result against the 0/23 contrast.
- **M2** (the dial): logistic validation bpc strictly improves along the whole ladder K = 16 → 4096, all four adjacent pairs (tie band ±0.005). Named alternative: non-monotone — as disclosed at registration, the pre-registration pilot run had hinted k* might dip at K=256, so cost might too.
- **MQ1** (the price): logistic validation bpc at K=4096 lands in [2.6519, 2.80] — within 0.15 bits of the per-coordinate quantization study's float32 reference (2.6519; quoted, not recomputed, and structurally not this dial's K→∞ limit, since a codebook-quantized reservoir only revisits stored anchors).

The registration also pre-stated the promotion rule: M1 firing at any K is provisional on a first seed; M1 *together with* MQ1 landing at the same K — a checkable automaton at negligible cost, the earlier quantization study's own named target — falls in the program's highest first-pass significance tier, which by standing policy requires a replication, queued ahead of all other pending work, before any entry in the program's confirmed-findings record. That is what happened. The seed-1 rerun registered the same three claims (N1/N2/NQ1), each with its own separately scored promotion rule.

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§4 Results

Table 1. Both registered passes (internal run log #16, seed 0; #17, seed 1; values seed 0 / seed 1). "Used" = distinct codewords active in the closed loop. Float32 reference: logistic 2.6519, ridge 3.1073 (per-coordinate quantization study, run log #7). No cell tripped the divergence tripwire — a fit-invalidating divergence rule — across 10/10 cells.

K	logistic val bpc	ridge val bpc	k*	used
16	3.7762 / 3.7695	4.0528 / 4.0668	8 / 7	16 / 15
64	3.3916 / 3.3665	3.7971 / 3.7706	8 / 7	63 / 64
256	3.1122 / 3.1085	3.5675 / 3.5644	7 / 10	252 / 252
1024	2.8401 / 2.8376	3.2742 / 3.2706	9 / 10	948 / 949
4096	**2.7089 / 2.7050**	**3.1406 / 3.1403**	**10 / 11**	**3320 / 3332**

M1/N1: 10 of 10 cells froze. Every codebook size, on both seeds, reached full suffix-determinism within a suffix depth of 7–11 characters — k*'s exact value is not seed-stable, but its finiteness is universal, always well inside the probe's depth limit of 20. The contrast with per-coordinate rounding is not a near-miss; it is total (0/23 there, 10/10 here). The freeze is not a collapse into a trivial handful of states: at K=4096, over 3,300 distinct codewords remain in active use.

M2/N2: the dial is smooth, 8/8. Logistic validation bpc improves strictly at every adjacent step of the ladder on both seeds, with gaps (0.385/0.279/0.272/0.131 at seed 0; 0.403/0.259/0.271/0.133 at seed 1) tens of times the tie band. The disclosed non-monotonicity concern did not materialize for cost, even though k* itself is mildly non-monotone in K at seed 0; the two curves do not have the same shape, which is recorded as an observation and not pursued further here.

MQ1/NQ1: the low-cost end of the dial compiles. K=4096 sits 0.0570 bits (seed 0) and 0.0531 bits (seed 1) above the float32 logistic reference — a third of the registered tolerance — while still freezing at k*=10/11. Ridge gives the same picture (3.1406/3.1403 against the reference's 3.1073), so the price is not an artifact of the trained instrument. Both halves of the result landed at the same K on both seeds; per the pre-stated rule, all three claims were promoted separately after the second pass.

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§5 Discussion

Cardinality, not resolution, is what compiles. Per-coordinate rounding makes the state grid coarser but leaves the reachable set effectively unbounded, and the rounding perturbation keeps the quantized system from becoming deterministic. A fixed codebook of visited states caps the reachable set at K outright — and at every tested K the closed-loop dynamics promptly become a function of a short window of recent input. The result parallels the automata-extraction literature, where finite abstractions of trained recurrent networks are constructed by external algorithms [2]; here the network *is* the automaton, exactly, by construction, and the question was only what that costs.

The cost is measured, small, and smooth. The compiled reservoir at K=4096 gives up about a twentieth of a bit of next-character prediction relative to its continuous counterpart, and the price rises predictably, without discontinuity, as the codebook shrinks toward 16. For this research program, an inspectable finite-state machine is therefore an object with a measured price rather than a hoped-for limit.

Open questions. Why should the codebook consist of *actually visited* states, as against an arbitrary codebook of the same size? Since this paper first appeared, the random-codebook control has been run, replicated across three seeds, and promoted: an arbitrary codebook compiles the reservoir only below K≈64, with K=64 itself a seed-sensitive boundary — see the companion study on random codebooks for the full result, including a first-pass threshold location that replication corrected. Also open: how far the near-free regime extends past K=4096, and whether the compiled automaton transfers across reservoirs.

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§6 References

1. R. M. Gray, "Vector quantization," *IEEE ASSP Magazine* 1(2), 1984.
2. G. Weiss, Y. Goldberg, E. Yahav, "Extracting automata from recurrent neural networks using queries and counterexamples," *ICML* 2018.
3. H. Jaeger, "The 'echo state' approach to analysing and training recurrent neural networks," GMD Report 148, German National Research Center for Information Technology, 2001.
4. M. Mahoney, "Large text compression benchmark" (text8), mattmahoney.net/dc/textdata.

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§7 Provenance

- **0c1eaba4** — `2026-08-11-caity-goat-3`: the design brief for this study; 5 cells, seed 0 (internal run log #16). Registered before running, with the pilot run disclosed (including that both pilot K values froze — flagged as suggestive only at that scale). The pre-stated highest-tier combination fired; the replication was queued ahead of all other pending work before any entry in the confirmed-findings record.
- **14080b1b** — `2026-08-11-goat-lattice-rerun1`: same 5-K sweep, seed 1 — an independent draw of the base reservoir, the warmup trajectory, and the codebook sample (internal run log #17). All three claims confirmed; promoted separately per the pre-stated rules.

All claims judged strictly against the registered wording; 2 seeds × 5 codebook sizes; replicated before publication. Internal designation: foothold lattice.
